JEPA-Contract: Joint-Embedding Prediction of Periodic Electronic-Structure Fields

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Abstract

Learning electronic structure directly from atomic geometry requires representations that connect periodic atomic configurations to both spatially resolved fields and global observables while respecting their distinct physical symmetries. We introduce **Periodic Field JEPA-Contract**, a joint-embedding predictive framework that maps periodic atomic graphs into a shared latent electronic representation from which electron density, electrostatic potential, and cohesive energy are inferred. The architecture combines periodic graph message passing, reciprocal-space field representations, analytical Fourier phases, and factorized Chebyshev Kolmogorov–Arnold Network transformations. A spectral field autoencoder first establishes the target electronic representation, after which a structure-conditioned predictor learns to infer this representation using only atomic species, fractional coordinates, and cell geometry. An exponential-moving-average target encoder stabilizes latent prediction, while field-specific decoders reconstruct the real-space observables and a translation-invariant spectral head predicts cohesive energy without requiring field reconstruction. We evaluate the framework using 30,000 converged density-functional-theory calculations for systematically deformed Al, Fe, and Ni periodic structures. We use Al and Fe for pretraining, whereas we use only **10%** of the Ni configurations to adapt to a chemically distinct domain. Periodic Field JEPA achieves **[0.0367]**, **[0.0296]**, and a cohesive-energy MAE of [**19.2 meV atom^{-1}**]. These results establish a modular approach for learning transferable, multi-observable electronic representations of periodic systems.

Keywords: JEPA, KAN, AI

Introduction

Interest in developing advanced materials has grown, driven by their wide range of applications, from optoelectronics[1–4] to nanomedicine and cancer treatment.[5,6] Advances in materials technology depend directly on a deeper understanding of how chemical and structural compositions govern electronic properties at atomic, molecular, and crystalline levels.[7] Against this backdrop, computational science plays a fundamental role in designing new materials by providing insights into relationships between the structural, electronic, and optoelectronic properties of material candidates that may be expensive to investigate experimentally.[8]

Nevertheless, traditional first-principles quantum chemistry calculations using wave function methodologies and density functional theory (DFT) still require high computational resources, limiting their application to systems with large chemical or configuration spaces. New machine learning (ML) algorithms have emerged as a complementary framework for electronic structure modeling, enabling the establishment of relationships between geometric

structure and molecular properties at reduced computational cost.[9,10]

In general, machine learning-based molecular frameworks bifurcate into two categories: molecular feature-based machine learning and end-to-end deep learning.[11] The latter encompasses different input representations, ranging from sequential data processed by Transformers, two-dimensional and volumetric representations employing convolutional neural networks, to structural representations via geometric deep learning. In this domain, architectures such as graph convolutional networks (GNNs), graph autoencoders, graph transformers, and message-passing models have been used to model drug design and nanocluster properties.[12–15] For atomic and periodic systems, the graph representation directly encodes geometric relations and periodic conditions, which provides a suitable structural representation. However, physical properties that differ in dimensionality and mathematical structure, such as high-dimensional electronic fields and global observables, may arise from a single atomic configuration, and learning only from the reconstruction of final outputs does not preserve information across different physical properties. From this perspective, a predictive objective defined in latent space provides a more suitable framework for learning relationships between geometric context and target properties.[16,17]

The Joint-Embedding Architecture (JEPA)[18] provides an appropriate formulation to predict a representation from latent space. In this architecture, the objective is to predict a latent representation of a target encoder from a context representation, rather than reconstruct the original space from final outputs. Studies in the literature indicate good performance of this framework applied to images with I-JEPA, videos with V-JEPA, three-dimensional point clouds with Point-JEPA, tabular data with T-JEPA, and multimodal alignment with M3-JEPA.[19–23] This reinforces the view that JEPA is a versatile framework for predictive relationships across different representations of the same underlying system. In addition, recent theoretical works demonstrate that predictions in latent space may favor informative factors while being less sensitive to irrelevant variations.[16,17]

For the JEPA framework, the predictor plays an important role in transforming the context data into a latent representation. In the present study, these transformations must capture the non-linearities between atomic structure and heterogeneous electronic properties. To increase the flexibility of the transformations, we parameterized both the encoder and the predictor using Kolmogorov–Arnold Networks (KANs), inspired by the Kolmogorov–Arnold representation theorem.[24]

The KAN formulation is an alternative to traditional multilayer perceptrons (MLP), where the fixed activation functions are replaced with trainable univariate functions. This new formulation has been applied across a broad range of fields, including differential-equation modeling, graph learning, human-activity recognition, and time-series forecasting.[24–28] A crucial component for this architecture is the choice of a univariate basis function: Wav-KAN employs wavelet functions to increase interpretability,[29] and Jacobi polynomial-based PointNet-KAN is suitable for shallower architectures.[30] Recent work shows that KAN modules incorporated into GNNs can perform well, replacing traditional MLP transformations in node embeddings, message passing, and readout.[31**?**,32] These results motivate us to use the KAN architecture as a predictor within the JEPA framework and related graph modules, where more flexible functions are needed to relate the shared latent representation to heterogeneous electronic properties.

In this work, we propose the Periodic Field Joint-Embedding Predictive Architecture (Periodic Field JEPA), a modular framework that combines graph-based encoders to incorporate conformational structure information with JEPA-based predictive learning in a shared latent space, as shown in Figure 1. From this representation, the model aims to provide a unified framework that connects periodic atomic structures with heterogeneous physical properties, including electron density, electrostatic potential, and energy. Additionally, we incorporate transformations based on KAN implementations with Chebyshev, Legendre, Hermite, and Laguerre basis functions into the predictor, message passing, and decoders.

To evaluate the hypotheses, we present extensive experiments across a fully converged dataset comprising 30,000 single-point quantum-chemistry calculations for periodic systems using the DFT method for three elements, aluminum (Al), iron (Fe), and nickel (Ni), including 10,000 configurations per element obtained from identical numerical settings.[33] The dataset includes structural representations alongside electron density, electrostatic potential grids, and energy. All cells share the same two-atom topology reference and undergo the same atomic displacements and cell deformations. All calculations in this dataset

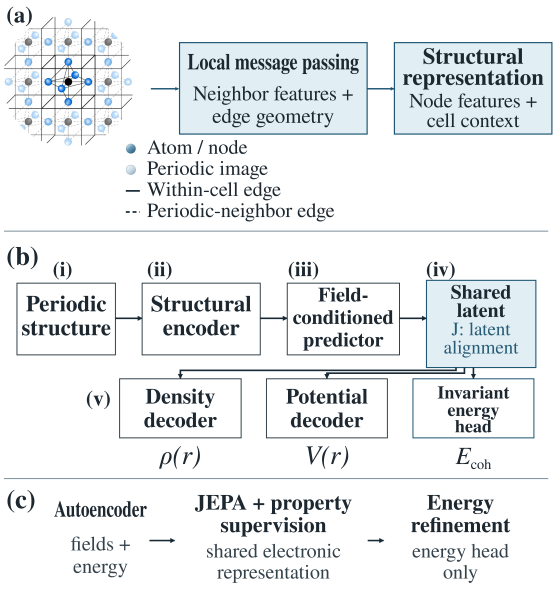

Figure 1. (a) Each periodic structure is represented as a graph. Node features describe the atoms and their local environments, while edge features describe interactions within the cell and across periodic images. Message passing combines this information with the cell geometry to obtain a structural representation. (b) The periodic structure (i) is processed by the structural encoder (ii) and the field-conditioned predictor (iii) to obtain a shared electronic latent representation (iv). Separate decoders use this representation to predict electron density and electrostatic potential, while an invariant head predicts cohesive energy (v). (c) Training starts with autoencoding of the two fields and auxiliary energy supervision. It then proceeds through JEPA alignment with progressively introduced property and periodicity objectives, followed by refinement of the energy head.

used a spin-unpolarized formalism; therefore, it should be interpreted as a highly controlled benchmark rather than a detailed study of these metals' thermodynamic or magnetic properties. For this study, we split the data as follows: Al and Fe for the pretraining step; Ni is treated as a chemically distinct transfer domain for the fine-tuning protocol, and only 10% of its configurations are used for adaptation.

Thus, this work presents a new JEPA-based modular architecture for predicting electronic properties from periodic structures. The principal contributions are combining graph-structure representation encoders with target-specific predictive representations, joint-embedding latent electronic representations, and different KAN-basis nonlinearities within a common framework for predicting electronic density, electrostatic potential, and energy.

Results

Joint-embedding predictive formulation

Our Periodic Field JEPA formulation aims to learn the connection between context structural periodic single-element atomic configuration and heterogeneous electronic properties: electronic density, electrostatic potential, and total energy. Therefore, the inference step receives only geometric and chemical information, and the reference fields and total energies are the targets for learning. Consequently, the electronic property outputs are inferred from structural representation encoded by a periodic graph.

In the adopted framework, we used two types of representations. Electron density and electrostatic potential are spatial fields, and we describe both using complex Fourier-coefficient features on the reciprocal lattice. In contrast, we predict the energy with a translation-invariant descriptor based on spectral magnitudes and correlations between electron-density and electrostatic-potential spectral components.

The Joint-Embedding Predictive Architectures (JEPAs) learn self-supervised representations by predicting the latent state of a target input from a context input, rather than reconstructing the original data. In this formulation, both the context and target encoders project the input data into an abstract latent space. At the same time, the predictor approximates the mapping from the context latent space to the target latent space. During JEPA training, the target encoder is updated via an Exponential Moving Average (EMA), preventing encoder collapse. The context encoder and predictor updates are made via backpropagation using gradient descent. This framework enables the model to extract semantically rich information from input data, abstracting away spurious features.

The architecture proposed in this work adopts a shared latent representation to connect the structural description of the periodic system with its electronic fields and total energy. In the JEPA encoder, structural and chemical information associated with the atoms, along with the periodic cell description, serves as input features to construct a latent representation of the atomic environment. This structural representation provides the context from which the predictor infers the system's electronic representation. The model uses a shared latent tensor for electron density, electrostatic potential, and total energy. The encoder and predictor

estimate the latent representation using only atomic and cell-geometry information. As a consequence, the predicted energy is conditioned on the potential and density fields. This formulation allows us to infer the fields and total energy from a common electronic representation and preserves the geometric symmetry associated with periodic systems.

Periodic GNN-KAN structural encoder

Once the latent space is structured, the next step is to determine non-linear transformations that connect the input features to the targets. In our framework, the KAN parametrization expands each input through a family of univariate basis functions and calculates each one's contribution to the output channels. In Periodic Field JEPA, this mechanism is used in message passing construction, node updates, spectral prediction, residual decoding, and the invariant energy head. The implementation supports Chebyshev, Legendre, and normalized Hermite polynomial families.

For this work, we selected the Chebyshev basis. This choice reflects its compatibility with the bounded KAN input domain, numerical stability via a three-term recurrence relation, and high approximation efficiency. It is important to distinguish the polynomial KAN basis from Fourier representations of electron density and electrostatic potential fields. Chebyshev parametrizes nonlinear transformations inside the neural network, while Fourier coefficients represent the fields. Figure 2 illustrates the KAN transformation adopted in this work.

Each atomic periodic configuration is represented by a graph, where the node features describe atomic species and chemical properties, whereas the edge features are the periodic displacements. In the dataset, fractional coordinates are used to describe the periodic image of each neighbor; the cell matrix then transforms these displacements into Cartesian vectors and interatomic distances. The set of periodic images is constructed from the instantaneous cell geometry, ensuring consistent graph connectivity under isotropic expansion, compression, anisotropic strain, and shear.

In the message-passing step, the node state is updated by incoming messages from neighboring nodes; the message is a function of the target node's current state, the relative difference across the edge, invariant edge attributes, and the global graph context. A global embedding is obtained by concatenating the final node representation with cell descriptors, which is used by the predictor to construct the shared latent representation of electron density and electronic potential. Node states are conditioned on Field-specific

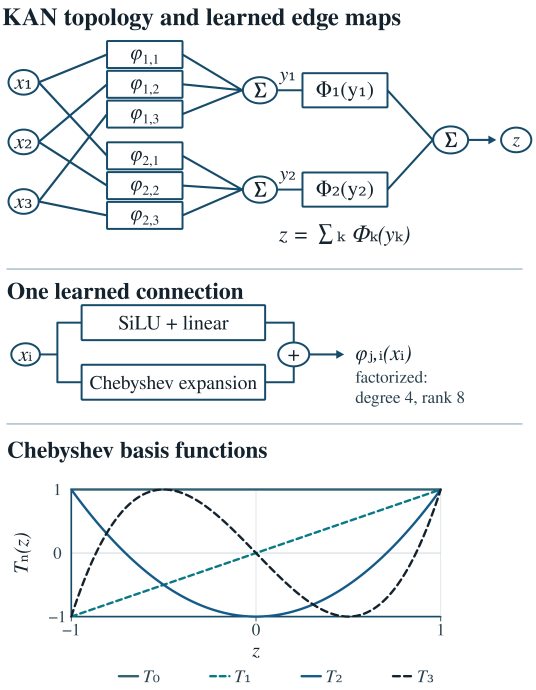

Figure 2. KAN topology and Chebyshev basis functions. Input features x_i are transformed by learned edge maps $\phi_{j,i}$ and aggregated to obtain the output z. Each learned connection combines a SiLU-linear pathway with a factorized Chebyshev expansion of degree $D = 4$ and rank $R = 8$. The lower plot shows the first four Chebyshev basis functions over $z \in [-1, 1]$.

queries, while directional message passing combined with reciprocal-mode embeddings identifies the target Fourier component. Analytical phases, determined by fractional atomic positions, are then combined with complex atomic amplitudes and aggregated over the reference cell. The predictor, field-specific residual decoders, and invariant energy head use independent instances of KAN-Chebyshev, sharing the same functional construction but with distinct learned weights.

Structural descriptors and property readouts

In our architecture, each node includes the atomic number, atomic mass, valence-electron count, periodic-table group, and period, along with a radial local descriptor, constructed by aggregating Gaussian radial basis function (RBF) embeddings of interatomic distances over the immediate neighborhood. For Ni transfer learning, the model initially starts from the Al–Fe pretrained model, with Ni-specific additive residual corrections initialized to zero and activated

only during the adaptation stages. Therefore, before Ni adaptation, its representation is mapped by physically pretrained features, while transfer learning only learns corrections to adapt the representation to Ni.

The edge features combine interatomic distance normalized by a cell-dependent radial scale, a logarithmic absolute-distance term, and Gaussian radial-basis response. We explicitly separate directional Cartesian vectors from scalar invariant edge features, reserving the directional vectors only when their information is required for field prediction. Global cell descriptors complement pooled atomic representations, with the full lattice matrix directly dictating real-space edge vectors and reciprocal-space wavevectors. As a consequence, variations in cell volume, anisotropy, and shear simultaneously modulate both the structural graph and the spectral electronic representation. Electron density and electrostatic potential fields are represented by aligned Fourier channels in a shared electronic latent space. Then, field-specific decoders subsequently reconstruct real-space fields via inverse Fourier transformation.

Cohesive energy is obtained from the same shared electronic representation used for electron density and electrostatic potential, without first reconstructing the real-space fields. Thus, it is important to note that cohesive energy must be invariant to rigid translations; therefore, the energy descriptor is constructed from the spectral magnitudes of the density and potential channels and their cross-channel correlations, for which the common translation-dependent phase cancels. Cohesive energy, electron density, and electrostatic potential are therefore formulated as distinct views of a single shared electronic potential.

Cohesive energy is predicted directly from the shared electronic representation without requiring real-space field reconstruction. To ensure translational invariance, the energy descriptor is constructed from the spectral magnitudes and cross-channel correlations of the density and potential fields, naturally canceling spatial phase shifts. Consequently, cohesive energy, electron density, and electrostatic potential serve as distinct observable views of a unified electronic potential."

Predictive Performance and Transferability

The Table1 shows the main metrics obtained for our architecture. For Al and Fe, the comparison between the ID and structural-OOD subsets shows only small variations after Ni adaptation. The same behavior is observed for Ni, for which the density relative-L2 remains approximately 0.047, the potential relative-L2 approximately 0.030, and the cohesive-energy MAE around to 0.012 eV/atom in both subsets. Therefore, the structural perturbations used to define the OOD evaluation do not produce a marked degradation of the final predictions. At the same time, the different behavior of the field errors and of the property-specific diagnostics indicates that the remaining deviations are not distributed uniformly over the electronic representation, but are concentrated in particular regions of the fields and of the cohesive-energy interval.

The Ni results provide a direct evaluation of the transfer protocol because this element is not included in the Al–Fe pretraining domain and only 10% of its configurations are used during adaptation. Before transfer learning, the Al–Fe pretrained model gives relative-L2 errors of 0.1399 and 0.2247 for the Ni density and electrostatic potential, respectively, together with a cohesive-energy MAE of 0.0371 eV/atom. After adaptation, these values decrease to 0.0474, 0.0306, and 0.0118 eV/atom for the Ni ID subset, which corresponds to error reductions of approximately 66%, 86%, and 68%. Similar reductions are obtained for the structural-OOD subset. This result is consistent with the adopted residual formulation: the pretrained representation already provides the initial physical mapping, while the Ni-specific terms introduce the corrections required to describe the new chemical domain without reconstructing the complete mapping from the beginning.

In addition, the adaptation to Ni does not increase the average errors obtained for Al and Fe. For Al ID, the density relative-L2 decreases from 0.0221 to 0.0176 and the potential relative-L2 from 0.0370 to 0.0321, while the cohesive-energy MAE changes from 0.0343 to 0.0316 eV/atom. For Fe ID, the corresponding values decrease from 0.0506 to 0.0451, from 0.0293 to 0.0262, and from 0.0133 to 0.0117 eV/atom. The same tendency is observed for the OOD subsets. Consequently, the information introduced during Ni adaptation is incorporated while the Al–Fe representation is retained, which is important for a shared model in which the same latent space is used to connect different chemical domains to electron density, electrostatic potential, and energy.

The cohesive-energy predictions in Figure 3 give us a complementary analysis of the shared electronic representation because the energy is obtained directly from the latent spectral descriptor. Fe and

Table 1. Summary metrics after Ni adaptation. Relative-L2 errors are reported for electron density and electrostatic potential.

Split	Density rel. L_2	Potential rel. L_2	High-ρ F1	V-sign F1
Al ID	0.0176	0.0321	0.8548	0.9964
Al OOD	0.0175	0.0318	0.8515	0.9965
Fe ID	0.0451	0.0262	0.9876	0.9966
Fe OOD	0.0451	0.0265	0.9873	0.9966
Ni ID	0.0474	0.0306	0.9921	0.9950
Ni OOD	0.0474	0.0304	0.9923	0.9950

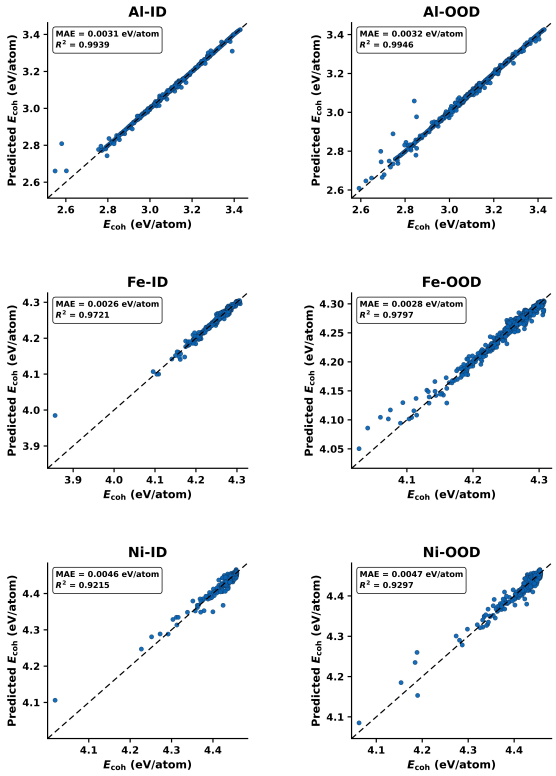

Figure 3. Cohesive-energy predictions for Al, Fe, and Ni under ID and structural-OOD evaluation, after Ni adaptation.

Ni predictions remain concentrated close to the DFT reference line for both ID and OOD subsets, with MAEs of 0.0117 and 0.0102 eV/atom for Fe and 0.0118 and 0.0124 eV/atom for Ni. Al follows the same global trend, but presents a broader residual distribution, with MAEs of 0.0316 eV/atom for ID and 0.0374 eV/atom for OOD. The signed residuals show that this difference is not associated with a single systematic displacement, since positive and negative errors occur over the sampled interval.

Table 2 shows the electron density confusion matrix results. Al has the smallest global density relative-L2 error, approximately 0.018, but its high-density F1 is 0.8548 for ID and 0.8515 for OOD. As shown in Figure 3, the model recovers about 81% of the reference high-density points for Al, while correctly classifying about 99% of points outside this region. Fe and Ni present higher global relative-L2 errors, close to 0.045 and 0.047, but recover the high-density region with F1 values close to 0.99. Consequently, the lower Al F1 does not contradict its smaller global reconstruction error. The relative-L2 norm is evaluated over the complete real space field, while the classification analysis isolates a restricted region in which local amplitude deviations have a larger contribution. The comparison therefore shows that field-average reconstruction and recovery of localized electronic regions provide complementary information about the learned representation.

Table 2. Row-normalized electron-density classification results. The column groups denote the reference class, and the subcolumns denote the predicted class.

	Ref. not high ρ		Ref. high ρ	
Partition	Pred. not high	Pred. high	Pred. not high	Pred. high
Al-ID	99.0	1.0	18.7	81.3
Al-OOD	99.0	1.0	19.3	80.7
Fe-ID	100.0	0.0	2.4	97.6
Fe-OOD	100.0	0.0	2.4	97.6
Ni-ID	99.9	0.1	1.1	98.9
Ni-OOD	99.9	0.1	1.1	98.9

Figure 4 shows the spatial color maps of the electron density. The density predictions reproduce the main spatial distribution observed in the reference maps for Al, Fe, and Ni. For Al, the high-density region around the center of the slice and the lower-density regions near the boundaries are preserved, while the residuals remain localized. The Fe prediction also recovers the main density features, although larger deviations are observed around the high-density region in the upper part of the map and in the central area. For Ni, the predicted field maintains the overall density profile, including the main high-density region, but shows smoother local variations than the reference. Overall, the results indicate that the model captures the dominant density patterns across the three elements,

while the remaining errors are spatially localized and depend on the material.

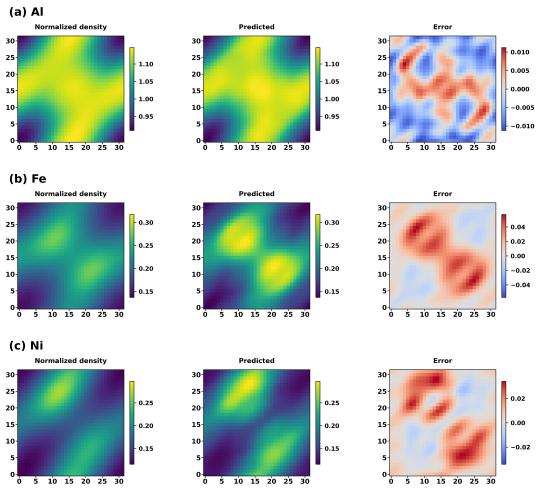

Figure 4. Electron Density maps for (a) Al, (b) Fe, and (c) Ni. For each material, the reference potential, model prediction, and spatial prediction error are shown.

Table 3 shows results for the electrostatic potential. After gauge centering, the potential-sign F1 remains between 0.9950 and 0.9966 for all evaluated subsets, with only small differences between ID and OOD. The confusion matrices show that both positive and non-positive regions are preserved with high agreement, even though the continuous potential relative-L2 varies between approximately 0.026 and 0.032. The sign analysis does not replace the continuous reconstruction error, but it evaluates whether the spatial separation of the gauge-centered potential is retained. Thus, the transfer step improves the field amplitude while maintaining the spatial organization associated with the sign of the electrostatic potential.

Figure 5 shows the spatial color maps of the electronic potential. The predicted electrostatic potential preserves the main spatial pattern of the reference map for Al, Fe, and Ni. For Al, the regions of higher and lower potential appear in similar positions, with localized residuals. The Fe prediction also captures the broad potential distribution, although its error map shows more pronounced differences around the central and lower-right regions. For Ni, the principal low-potential regions remain identifiable, while the residuals form a structured pattern across the slice. Together, the panels show that the model recovers

Table 3. Row-normalized electrostatic-potential sign classification results. The column groups denote the reference sign, and the subcolumns denote the predicted sign.

	Ref. $V \le 0$		Ref. $V > 0$	
Partition	Pred. $V \le 0$	Pred. $V > 0$	Pred. $V \le 0$	Pred. $V > 0$
Al-ID	98.4	1.6	0.2	99.8
Al-OOD	98.5	1.5	0.2	99.8
Fe-ID	98.7	1.3	0.3	99.7
Fe-OOD	98.7	1.3	0.3	99.7
Ni-ID	98.6	1.4	0.6	99.4
Ni-OOD	98.6	1.4	0.6	99.4

the overall potential distribution while retaining local, element-dependent errors.

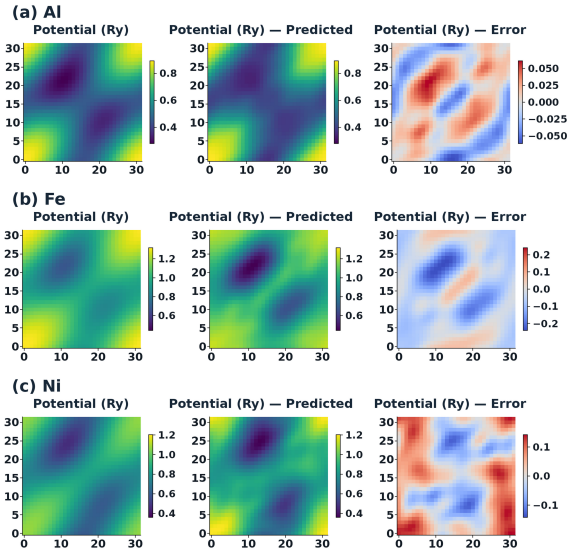

Figure 5. Electrostatic potential maps for (a) Al, (b) Fe, and (c) Ni. For each material, the reference potential, model prediction, and spatial prediction error are shown. The color bars report potential and error values in Ry.

Concluding Remarks

Methods

Periodic graph representation

In our framework, a graph representation $G = (V, E)$ is used to described single specie periodic atomic configuration. For this approximation, the nodes correspond to atomic centers and the edges to interactions between atoms in the reference cell

and their periodic images. To preserve the stable neighborhood topology under cell deformations while allowing the physical edge geometry to vary, we define the periodic neighborhood from fractional atomic displacements and then obtain the corresponding Cartesian geometry from the instantaneous cell matrix. Since the complete cell matrix transforms each retained fractional displacement into Cartesian space, changes in the periodic cell modify the corresponding interatomic vectors and distances. As a consequence, isotropic expansion, compression, anisotropic strain, and shear are naturally incorporated into the structural representation without requiring the graph connectivity to be redefined in Cartesian space.

$$\mathbf{q}_{ij}^{(\mathbf{n})} = \mathbf{s}_j + \mathbf{n} - \mathbf{s}_i, \qquad \mathbf{r}_{ij}^{(\mathbf{n})} = \mathbf{q}_{ij}^{(\mathbf{n})}\mathbf{A}, \tag{1}$$

where $\mathbf{s}_i$ and $\mathbf{s}_j$ represent the fractional coordinates of atoms i and j, respectively, $\mathbf{n} \in \mathbf{Z}^3$ is an integer lattice-image shift, and $\mathbf{A}$ is the simulation-cell matrix. The periodic edge is retained when the fractional displacement satisfies the following condition

$$\left\|\mathbf{q}_{ij}^{(\mathbf{n})}\right\|_\infty \le \varepsilon, \tag{2}$$

and zero-distance self-interactions are excluded.

In our architecture, each node contains physical descriptors associated with the atomic species, including atomic number, atomic mass, valence-electron count, periodic-table group, and period. In a complementary manner, the local periodic atomic environment is encoded using Gaussian radial basis functions (RBFs) aggregated over the periodic neighborhood. The radial basis response is defined as

$$\phi_m(d) = \exp\left[-\frac{1}{2}\left(\frac{d-\mu_m}{\sigma}\right)^2\right], \tag{3}$$

where the Cartesian interatomic distance is associated with the periodic edge, μ_m denotes the center of the m-th Gaussian radial basis function, σ defines its width, and m indexes the set of radial basis functions.

The edge features include the interatomic distances normalized by a cell-dependent radial scale, a logarithmic term, and the corresponding Gaussian radial basis responses. Additionally, we keep Cartesian directions separated from these invariant descriptors, and this directional information is used for the field-conditioned prediction stage, where the spatial orientation of the electronic fields becomes relevant. At the global level, the pooled atomic representation is complemented by descriptors of the periodic cell, including the lattice-vector lengths, inter-vector angle cosines, logarithmic volume per atom, and normalized atom count.

Factorized Chebyshev KAN parameterization

Once the structural representation is defined, the next step is to define the nonlinear transformations that connect the atomic information to the target electronic representation. In the Periodic Field JEPA architecture, these transformations are parameterized using factorized polynomial Kolmogorov Arnold Network (KAN) layers. KAN transformations are used in the message-passing blocks, spectral predictor, field-specific residual decoders, and cohesive-energy head.

Our implementation allows us to employ different polynomial basis families for KAN expansion. For this work, we selected the Chebyshev basis because it provides a stable recursive construction over the bounded input domain adopted by the KAN layers. Before the polynomial expansion, each input component can be written according to

$$z_i = \tanh(x_i), \tag{4}$$

and the Chebyshev basis is generated through the recurrence relation

$$\begin{aligned} T_0(z_i) &= 1, \qquad T_1(z_i) = z_i, \\ T_d(z_i) &= 2z_i T_{d-1}(z_i) - T_{d-2}(z_i). \end{aligned} \tag{5}$$

A dense polynomial coefficient tensor would introduce a huge number of parameters across the input channels, output channels, and polynomial order. Therefore, we use a factorized form of the KAN transformation.

$$y_o = \left[\mathbf{W}_{\text{base}}\,\text{SiLU}(\mathbf{x})\right]_o + \sum_{r=1}^{R} A_{or} \sum_i \sum_{d=1}^{D} B_{rid} T_d(z_i). \tag{6}$$

In the above expression, y_o is the o-th output feature, x is the input vector, and W_{base} contains the weights of the SiLU-linear branch. A_{or} projects the r-th low-rank component onto output o, while B_{rid} weights the contribution of the d-th Chebyshev polynomial $T_d(z_i)$ associated with input feature i. In this work, $D = 4$ is the polynomial degree, and $R = 8$ is the factorization rank.

Periodic message passing and structural context

The elemental and local site descriptors are first embedded to initialize the atomic node representations, and then they are processed by four periodic message-passing blocks. This stage is important to aggregate information from each atomic environment with information from the complete periodic configuration.

At each layer l, the node states are normalized, and a global graph context is obtained by applying mean pooling,

$$\mathbf{g}^{(l)} = \frac{1}{|V|} \sum_{i \in V} \widetilde{\mathbf{h}}_i^{(l)}. \tag{7}$$

For each node update $j \to i$, the message passing depends on the current state of the receiving node, the relative difference between the neighboring and receiving node states, the invariant edge attributes, and the global graph context,

$$\mathbf{m}_{ij}^{(l)} = \mathcal{K}_m^{(l)} \left(\widetilde{\mathbf{h}}_i^{(l)} \oplus \left[\widetilde{\mathbf{h}}_j^{(l)} - \widetilde{\mathbf{h}}_i^{(l)} \right] \oplus \mathbf{e}_{ij} \oplus \mathbf{g}^{(l)} \right), \tag{8}$$

where $\oplus$ denotes concatenation and $\mathcal{K}_m^{(l)}$ represents the KAN transformation used for message passing step.

The incoming messages are averaged over the periodic neighborhood,

$$\overline{\mathbf{m}}_i^{(l)} = \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} \mathbf{m}_{ij}^{(l)}, \tag{9}$$

and then are incorporated into the node representation,

$$\mathbf{h}_i^{(l+1)} = \mathbf{h}_i^{(l)} + \sigma(\gamma_l)\, \mathcal{K}_u^{(l)} \left(\widetilde{\mathbf{h}}_i^{(l)} \oplus \overline{\mathbf{m}}_i^{(l)} \oplus \mathbf{g}^{(l)} \right). \tag{10}$$

The learned gate $\sigma(\gamma_l)$ controls the contribution introduced by each update. After the final message-passing block, the atom-resolved representations are mean-pooled and combined with the cell descriptors. This operation defines the global structural context that the predictor subsequently uses to infer the electronic representation.

As the Ni fine-tuning stage starts from the Al–Fe pretrained representation, Ni-specific additive corrections are added to the atomic embedding. Therefore, before the fine-tuning step, the Ni representation is determined by the same physical descriptors and transformations learned during Al–Fe pretraining, and in the transfer-learning stage, the architecture learns only the residual corrections required to adapt this representation to Ni.

Spectral representation of electron density and electrostatic potential

Electron density and electrostatic potential are defined as fields over the periodic cell. In our framework, both electronic properties are represented on a canonical three-dimensional grid and transformed to reciprocal space using the Fourier transform.

Before applying the Fourier transform, we normalize the electron density using the cell volume and the number of valence electrons for each chemical element, whereas the electrostatic potential is centered with respect to its spatial mean and scaled using statistics determined exclusively from the permitted pretraining data,

$$\widetilde{\rho}(\mathbf{r}) = \rho(\mathbf{r}) \frac{\Omega}{N_e}, \qquad \widetilde{V}(\mathbf{r}) = \frac{V(\mathbf{r}) - \langle V \rangle}{s_V}, \tag{11}$$

where Ω is the cell volume, N_e is the number of valence electrons, and s_V is the normalization scale used for the electrostatic potential.

For each electron density and electrostatic potential field, we used 512 independent modes. The real and imaginary components of the Fourier coefficients are represented separately, which gives us 1024 real components for each field channel. Therefore, electron density and electrostatic potential occupy aligned channels associated with the same reciprocal modes.

Spectral field autoencoder

Before starting the JEPA training, we first train a spectral field autoencoder using electron density and electrostatic potential fields. The purpose of this stage is to obtain an appropriate electronic latent representation before learning the mapping from atomic structure to electronic properties.

The encoder stage operates on the normalized Fourier coefficients and applies a phase-equivariant residual transformation. This transformation allows the density and potential channels associated with the same reciprocal mode to exchange information while preserving their individual channels and complex phase structure.

The reconstruction decoder maps the latent coefficients back to the canonical spectral grid, followed by an inverse real Fourier transform. For the density branch, the reconstruction is bounded in

the logarithmic representation, constrained to positive values, and renormalized. For the electrostatic potential, an amplitude correction is applied, and the spatial mean is removed to preserve a consistent potential gauge.

The loss autoencoder is calculated according to

$$\mathcal{L}_{\mathrm{AE}} = \mathcal{L}_\rho + \mathcal{L}_V + \lambda_{\mathrm{anchor}}\mathcal{L}_{\mathrm{anchor}} + \lambda_{\mathrm{coh}}\mathcal{L}_{\mathrm{coh}}, \quad (12)$$

where $\mathcal{L}_\rho$ and $\mathcal{L}_V$ correspond to the electron density and electrostatic potential reconstruction losses, respectively. The latent anchor provides an additional constraint on the electronic representation. It is important to note that cohesive energy is neither an input to the autoencoder nor a field reconstructed by the decoder. Instead, cohesive energy is predicted from the shared electronic latent representation. The resulting prediction error provides auxiliary supervision that encourages this representation to retain information relevant to energy.

After selecting the best autoencoder checkpoint, we freeze the original field encoder and reconstruction decoder. Copies of the encoder weights initialize the predictor's spectral transformation and the electronic online and target encoders used in the subsequent JEPA training.

Field-conditioned structural predictor

After defining the initial latent electronic representation using the field autoencoder, the next step is to learn how this representation can be inferred from the periodic atomic structure information. For this purpose, the structural predictor receives only atomic and cell information and maps the periodic graph to the electronic spectral representation.

For each field query, the atom-resolved structural states are combined with the global graph context and processed through an additional directional message-passing operation. The resulting density and potential tokens subsequently interact through self-attention. Therefore, information associated with one electronic channel can contribute to the prediction of the other before the reciprocal-space coefficients are generated.

For each atom and reciprocal mode, the predictor combines the field-conditioned node state with a learned reciprocal-mode embedding, the reciprocal-index direction, and a cell-dependent wavevector magnitude. A KAN transformation followed by a linear readout generates a complex atomic amplitude.

The atomic positions enter the spectral construction through the Fourier phase,

$$\varphi_{i\mathbf{k}} = \exp\left(-2\pi i\, \mathbf{k}\cdot\mathbf{s}_i\right), \quad (13)$$

and the corresponding atomic contributions are subsequently averaged over the reference cell,

$$\widehat{C}_{f\mathbf{k}} = \frac{1}{N}\sum_{i=1}^{N} A_{if\mathbf{k}}\,\varphi_{i\mathbf{k}}, \quad (14)$$

here, $\widehat{C}_{f\mathbf{k}}$ is the predicted Fourier coefficient of field f at reciprocal mode $\mathbf{k}$; N is the number of atoms in the reference cell; i indexes these atoms; and $A_{if\mathbf{k}}$ is the complex amplitude predicted for atom i, field f, and mode $\mathbf{k}$. The field index $f \in \{\rho, V\}$ refers to electron density or electrostatic potential. Division by N normalizes the sum, while the phases determine whether atomic contributions reinforce or cancel at each mode.

The predicted Fourier coefficients of density and potential occupy aligned channels of the same electronic representation. In this way, the two electronic fields can exchange information in reciprocal space while remaining distinguishable channels of the common representation.

JEPA target representation and variable EMA

In this work, the JEPA formulation is used to connect the periodic structural latent representation to the electronic latent representation defined by the field autoencoder. At the beginning of JEPA training, the selected autoencoder encoder initializes three independent parameter sets: the spectral transformation used by the structural predictor, an electronic online encoder, and a target encoder. The original autoencoder remains frozen during JEPA training and transfer learning.

During training, the target encoder receives the reference electron density and electrostatic potential, whereas the structural branch receives only the atomic and cell information. The role of the structural predictor is therefore to infer, from the periodic graph, an electronic latent state that matches the representation provided by the electronic target branch.

The target encoder does not receive gradients from backpropagation updates. Its parameters are updated from the electronic online encoder using an exponential moving average (EMA),

$$\boldsymbol{\theta}_{\mathrm{target}}^{(t+1)} = \tau_t \boldsymbol{\theta}_{\mathrm{target}}^{(t)} + (1-\tau_t)\,\boldsymbol{\theta}_{\mathrm{electronic}}^{(t+1)}. \quad (15)$$

For this work, the EMA coefficient is not kept fixed during optimization. Instead, τ is progressively increased from $\tau_{\min} = 0.994$ to $\tau_{\max} = 0.998$ using a cosine schedule,

$$\tau(p) = \tau_{\min} + \frac{\tau_{\max} - \tau_{\min}}{2} \left[1 - \cos(\pi p)\right], \quad (16)$$

where $p \in [0, 1]$ represents the normalized progress across the training curriculum.

At the beginning of training, the lower value of τ allows the target representation to respond more rapidly to changes in the electronic online encoder. As optimization progresses, the larger value increases the temporal memory of the target encoder and produces a progressively more stable latent target. The EMA schedule is coordinated across the curriculum and is not restarted from its minimum value at each training stage.

During the initial latent-alignment stage, the electronic online encoder is also kept close to the pretrained field representation and its contribution is progressively released. This procedure reduces the movement of the electronic target while the structural predictor establishes the initial mapping between the periodic graph and the electronic latent representation.

Field reconstruction and cohesive-energy readout

The predicted latent space, obtained from the periodic structure, is used to predict electron density, electrostatic potential, and cohesive energy. However, these properties require different readout operations: the first two are spatial fields, while cohesive energy is a global scalar observable.

For electron density and electrostatic potential, the corresponding latent channels are processed by separate predictive decoders. Each decoder contains a spectral basis mapping initialized from the pretrained autoencoder reconstruction decoder, plus a nonlinear residual correction. The resulting spectral coefficients are transformed back to real space using an inverse real Fourier transform.

Cohesive energy is obtained from the same electronic latent representation of electronic density and electrostatic potential. Since cohesive energy is a global scalar, it must be invariant to translations of the periodic structure. A rigid translation changes the phases of the complex spectral coefficients but not their magnitudes. The energy descriptor is therefore constructed from spectral magnitudes and density-potential cross-correlations, for which the phase factors cancel.

For each reciprocal-space mode $\mathbf{k}$, the corresponding density and potential coefficients are written as

$$C_{\rho,\mathbf{k}} = r^R_{\mathbf{k}} + i r^I_{\mathbf{k}}, \qquad C_{V,\mathbf{k}} = v^R_{\mathbf{k}} + i v^I_{\mathbf{k}}, \quad (17)$$

where $\mathbf{k}$ denotes the reciprocal-space mode, i is the imaginary unit, and the superscripts R and I identify the real and imaginary components of each coefficient. The spectral magnitude of a coefficient is then obtained as

$$a_{\rho,\mathbf{k}} = \left|C_{\rho,\mathbf{k}}\right|, \qquad a_{V,\mathbf{k}} = \left|C_{V,\mathbf{k}}\right|. \quad (18)$$

In addition to the individual spectral amplitudes, the energy descriptor contains information about the relationship between the density and potential channels. The real and imaginary components of their cross-spectrum can be written as

$$c^R_{\mathbf{k}} = \frac{r^R_{\mathbf{k}} v^R_{\mathbf{k}} + r^I_{\mathbf{k}} v^I_{\mathbf{k}}}{1 + a_{\rho,\mathbf{k}} a_{V,\mathbf{k}}}, \qquad c^I_{\mathbf{k}} = \frac{r^I_{\mathbf{k}} v^R_{\mathbf{k}} - r^R_{\mathbf{k}} v^I_{\mathbf{k}}}{1 + a_{\rho,\mathbf{k}} a_{V,\mathbf{k}}}. \quad (19)$$

A rigid translation of the periodic structure changes the phase of each reciprocal-space coefficient in the same way for both electron density and electrostatic potential fields. This phase change does not affect the spectral magnitudes. It also cancels when density and potential coefficients are combined through cross-channel products. Therefore, these quantities represent the electronic fields in a way that is invariant to translations of the periodic system. The cohesive-energy descriptor is then written as

$$\mathbf{d}_E = \left[\log\left(1 + a_{\rho,\mathbf{k}}\right), \log\left(1 + a_{V,\mathbf{k}}\right), c^R_{\mathbf{k}}, c^I_{\mathbf{k}}\right]_{\mathbf{k}\neq\mathbf{0}}. \quad (20)$$

The logarithmic terms reduce the scale variation among spectral modes. After normalization and projection, $\mathbf{d}_E$ is processed by an independent KAN–Chebyshev transformation and a scalar readout to predict the cohesive energy. Thus, the density, potential, and energy predictions are predicted from the same latent electronic representation. The energy branch does not use an additional latent representation or a direct bypass from the structural context.

Cohesive-energy target and property objectives

In this work, we used the cohesive energy as the scalar target. For each element X, the cohesive energy is defined as

$$E_{\mathrm{coh}}^{(X)} = E_{\mathrm{anchor}}^{(X)} + E_{\mathrm{ref}}^{(X)} - \frac{E_{\mathrm{DFT}}}{N}, \tag{21}$$

where $E_{\mathrm{anchor}}^{(X)}$ is an element-dependent thermochemical reference, $E_{\mathrm{ref}}^{(X)}$ is the corresponding bulk reference estimated only from the allowed training configurations, and E_{DFT}/N is the total DFT energy divided by the number of atoms N in the periodic structure.

The normalization of cohesive energy is obtained only from the pretraining data using center and scale statistics. In this way, information from evaluation subsets does not enter the definition of the scalar target.

It is important to highlight that electron density, electrostatic potential, and cohesive energy have different physical characteristics; each property is optimized using an objective adapted to its representation. For electron density, the predicted field $\hat{\rho}(\mathbf{r})$ is compared with the DFT reference $\rho(\mathbf{r})$ in both real and reciprocal space. The real-space terms constrain the density values and their Cartesian spatial variations on the grid, whereas the spectral term compares the Fourier coefficients.

$$\mathcal{L}_{\rho} = 0.33\,\mathcal{L}_{\log\rho} + 0.50\mathcal{L}_{\mathrm{rel}} + 0.12\mathcal{L}_{\nabla} + 0.05\mathcal{L}_{\mathrm{spec}}. \tag{22}$$

In this equation, $\mathcal{L}_{\log\rho}$ is the Smooth L1 error between the logarithms of the predicted and reference densities. $\mathcal{L}_{\mathrm{rel}}$ is the relative L_2 error of the full density field. $\mathcal{L}_{\nabla}$ compares their Cartesian gradients and constrains local spatial variations. Finally, $\mathcal{L}_{\mathrm{spec}}$ compares their reciprocal-space coefficients, with increased weight for higher-frequency modes. These terms supervise the density decoder and constrain the shared electronic representation.

For the electrostatic potential, the predicted and reference fields are first centered to remove the arbitrary additive gauge. The objective then combines complementary terms in real and reciprocal space,

$$\begin{aligned}\mathcal{L}_{V} = {} & 0.15\,\mathcal{L}_{\mathrm{shape}} + 0.25\,\mathcal{L}_{\mathrm{rel}} + 0.15\,\mathcal{L}_{\mathrm{amp}} \\ & + 0.10\,\mathcal{L}_{\cos} + 0.20\,\mathcal{L}_{\nabla} + 0.15\,\mathcal{L}_{\mathrm{spec}}.\end{aligned} \tag{23}$$

In the above equation, $\mathcal{L}_{\mathrm{shape}}$ is the normalized Smooth L1 error between the two potential fields, whereas $\mathcal{L}_{\mathrm{rel}}$ measures their relative L_2 error. The amplitude term $\mathcal{L}_{\mathrm{amp}}$ constrains the root-mean-square magnitude of the potential, and $\mathcal{L}_{\cos}$ measures the agreement between the full predicted and reference fields. The terms $\mathcal{L}_{\nabla}$ and $\mathcal{L}_{\mathrm{spec}}$ compare, respectively, the Cartesian gradients and the Fourier coefficients of the two fields. Together, these terms supervise the potential decoder and constrain the shared electronic representation.

For the cohesive-energy head, the regression objective combines Smooth-L_1 and mean-squared errors,

$$\mathcal{L}_{\mathrm{coh}} = 0.65\,\mathcal{L}_{\mathrm{SmoothL1}} + 0.35$$

The Smooth-L_1 term provides a robust error measure, whereas the mean-squared term increases the penalty assigned to larger deviations. Configurations in the weak-cohesion region receive larger sample weights, with an additional increase for the extreme lower tail of the cohesive-energy distribution. The loss also assigns a larger penalty when the predicted cohesive energy exceeds the reference value, since this corresponds to an overestimation of the binding strength.

Progressive curriculum and optimization protocol

The different tasks considered in our architecture introduce objectives with different levels of complexity. Therefore, instead of introducing all losses at the beginning of training, we use a progressive curriculum in which new electronic and physical constraints are introduced gradually. As can be observed in Figure 6.